Formally, we define

$$\mathbf{Q}^2 \stackrel{\text{def}}{=} \begin{array}{c} \begin{array}{ccccc} \cdots & \overbrace{}^{\frac{c(t)}{t+1}} & \cdots & \overbrace{}^{\frac{1}{t+1}} & \overbrace{}^{\frac{1}{(t+1)^2}} & \cdots \end{array} \\ \begin{pmatrix} & 1 & & & & \\ & & & 1 & & \\ & & & & \frac{1}{3} & \end{pmatrix} \end{array} \tag{103a}$$

$$\mathbf{K}^2 \stackrel{\text{def}}{=} \begin{array}{c} \begin{array}{ccccc} \cdots & \overbrace{}^{\frac{c(t-1)}{t+1}} & \cdots & \overbrace{}^{\frac{1}{t+1}} & \overbrace{}^{\frac{1}{(t+1)^2}} & \cdots \end{array} \\ \begin{pmatrix} & & & 1 & & \\ & -1 & & & & \\ & & & & \frac{1}{3} & \end{pmatrix} \end{array} \tag{103b}$$

$$f(\mathbf{q}, \mathbf{k}) \stackrel{\text{def}}{=} -|\langle \mathbf{q}, \mathbf{k} \rangle| \tag{103c}$$

$$\mathbf{V}^2 \stackrel{\text{def}}{=} \begin{array}{cc} \begin{array}{ccccc} \cdots & \overbrace{}^{[\![v_{t-1}]\!]} & \cdots & \overbrace{}^{t+1} & \cdots \end{array} & \\ \begin{pmatrix} & & & & \\ & \mathbf{I}_{|\Gamma|} & & 1 & \\ & & & & \end{pmatrix} & \begin{array}{c} \vdots \\ \ell(t)+1 \\ [\![v_{\ell(t)}]\!] \\ \vdots \end{array} \end{array} \tag{103d}$$

$$\mathbf{O}^1 = \mathbf{0}_{D \times D} \tag{103e}$$

Pérez et al. (2021, Lemma 10) show that, given the parameters above, the output of the scoring function f is maximized at the entry $\ell(t) \in \{0, \ldots, t\}$.[35] Since $\mathbf{V}^2$ copies the second value from the positional encoding of the symbol at time step $\ell(t)$, which is $\ell(t)+1$, this entry appears in the output of the attention mechanism. Furthermore, $\mathbf{V}^2$ also copies the value $[\![v_{\ell(t)}]\!]$ from the same entry. This, together with the zero-valued function O and residual connections, results in Eq. (102). ■

Lemma F.8 (Correctness of the output function)**.** *There exists an MLP F that, given the outputs $\mathbf{X}^2$ of $\mathcal{L}_2$, computes the one-hot encoding of the current configuration of the PTM. More concretely, it holds that*

$$F\left(\mathbf{x}_t^2\right) = [\![q_t, s_t, a_{t-1}]\!]. \tag{104}$$

Proof. Here, we define a function F similar to that of Pérez et al. (2021, Lemma 11), but with an additional layer that handles the addition of the $\perp$ symbol not handled by Pérez et al. (2021). The logic nonetheless remains the same: F receives the output of $\mathcal{L}_2$ of the form

$$\mathbf{x}_t^2 = \begin{pmatrix} [\![q_t]\!] \\ [\![v_{t-1}]\!] \\ [\![a_{t-1}]\!] \\ a_{t-1}, a_{t-2} \\ \frac{c(t)}{t+1}, \frac{c(t-1)}{t+1} \\ \ell(t)+1, [\![v_{\ell(t)}]\!] \\ 1, t+1, \frac{1}{t+1}, \frac{1}{(t+1)^2} \end{pmatrix}. \tag{105}$$

and *(1)* copies the value of q_t, *(2)* copies the value of a_{t-1} *(3)* compares the value of $\ell(t)$ to t to determine whether $s_t = \sqcup$ or $s_t = v_{\ell(t)}$, *(4)* compares the value of t to 0 to determine whether $s_t = \perp$ or $s_t = v_{\ell(t)}$.

[35] More precisely, their construction results in the maximum being at $\ell(t+1)$, since layer 1 in their construction, in contrast to ours, computes the values $\frac{c(t+1)}{t+1}$ and $\frac{c(t)}{t+1}$, shifting all computations by one step. This is the consequence of the aforementioned difference between their deterministic and our probabilistic framework.

Concretely, F will take the form of a three-layer MLP

$$F(\mathbf{x}) \stackrel{\text{def}}{=} \mathrm{ReLU}(\mathbf{W}_5\mathrm{ReLU}(\mathbf{W}_4\mathrm{ReLU}(\mathbf{W}_3\mathrm{ReLU}(\mathbf{W}_2\mathrm{ReLU}(\mathbf{W}_1\mathbf{x}+\mathbf{b}_1)+\mathbf{b}_2)+\mathbf{b}_3)+\mathbf{b}_4)+\mathbf{b}_5) \tag{106}$$

where

$$\mathbf{W}^1 \in \mathbb{R}^{(|Q|+|\Gamma|+|A|+|\Gamma|+|\Gamma|+2)\times D} \tag{107a}$$
$$\mathbf{W}^2 \in \mathbb{R}^{(|Q|+|\Gamma|+|A|+|\Gamma|+|\Gamma|+2)\times(|Q|+|\Gamma|+|A|+|\Gamma|+|\Gamma|+2)} \tag{107b}$$
$$\mathbf{W}^3 \in \mathbb{R}^{(|Q|+|\Gamma|+|A|+|\Gamma|+|\Gamma|)\times(|Q|+|\Gamma|+|A|+|\Gamma|+|\Gamma|+2)} \tag{107c}$$
$$\mathbf{W}^4 \in \mathbb{R}^{(|Q|+|\Gamma|+|A|)\times(|Q|+|\Gamma|+|A|+|\Gamma|+|\Gamma|)} \tag{107d}$$
$$\mathbf{W}^5 \in \mathbb{R}^{(|Q||\Gamma||A|)\times(|Q|+|\Gamma|+|A|)} \tag{107e}$$
$$\tag{107f}$$

are the weights of the MLP and $\mathbf{b}^1, \ldots, \mathbf{b}^5$ are the corresponding biases. For conciseness, we will denote

$$C_1 \stackrel{\text{def}}{=} |Q| \tag{108a}$$
$$C_2 \stackrel{\text{def}}{=} C_1 + |\Gamma| \tag{108b}$$
$$C_3 \stackrel{\text{def}}{=} C_2 + |A| \tag{108c}$$
$$C_4 \stackrel{\text{def}}{=} C_3 + |\Gamma| \tag{108d}$$
$$C_5 \stackrel{\text{def}}{=} C_4 + |\Gamma| \tag{108e}$$
$$C_6 \stackrel{\text{def}}{=} C_5 + 2 \tag{108f}$$

and

$$D_1 \stackrel{\text{def}}{=} |Q| \tag{109a}$$
$$D_2 \stackrel{\text{def}}{=} D_1 + |\Gamma| \tag{109b}$$
$$D_3 \stackrel{\text{def}}{=} D_2 + |A| \tag{109c}$$
$$D_4 \stackrel{\text{def}}{=} D_3 + 2 \tag{109d}$$
$$D_5 \stackrel{\text{def}}{=} D_4 + 2 \tag{109e}$$
$$D_6 \stackrel{\text{def}}{=} D_5 + 1 + |\Gamma| \tag{109f}$$
$$D_7 \stackrel{\text{def}}{=} D_6 + 4 \tag{109g}$$

Then, we define

$$\mathbf{W}^1 = \begin{array}{c} \begin{array}{ccccccc} \overbrace{\quad}^{1:D_1} & \overbrace{\quad}^{D_1:D_2} & \overbrace{\quad}^{D_2:D_3} & \cdots & \overbrace{\quad}^{D_5+1} & \overbrace{\quad}^{D_6+1} & \overbrace{\quad}^{D_6+2} & \cdots \end{array} \\ \begin{pmatrix} \mathbf{I}_{|Q|} & & & & & & & \\ & \mathbf{I}_{|\Gamma|} & & & & & & \\ & & \mathbf{I}_{|A|} & & & & & \\ & & & & & & & \\ & & & & 1 & 1 & -1 & \\ & & & & & 2 & -1 & \end{pmatrix} \end{array} \qquad \mathbf{b}^1 = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ [\![\sqcup]\!] \\ [\![\perp]\!] \\ 0 \\ 0 \end{pmatrix} \tag{111}$$

$$\tag{110}$$

$$\mathbf{W}^2 = \begin{array}{c} \begin{array}{ccccccc} \overbrace{\phantom{\mathbf{I}_{|Q|}}}^{1:C_1} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_1:C_2} & \overbrace{\phantom{\mathbf{I}_{|A|}}}^{C_2:C_3} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_3:C_4} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_4:C_5} & \overbrace{}^{C_5+1} & \overbrace{}^{C_5+2} \end{array} \\ \begin{pmatrix} \mathbf{I}_{|Q|} & & & & & & \\ & \mathbf{I}_{|\Gamma|} & & & & & \\ & & \mathbf{I}_{|A|} & & & & \\ & & & \mathbf{I}_{|\Gamma|} & & & \\ & & & & \mathbf{I}_{|\Gamma|} & & \\ & & & & & 1 & -1 \\ & & & & & & 1 \end{pmatrix} \end{array} \quad (112) \qquad \mathbf{b}^2 = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|\Gamma|} \\ 0 \\ 0 \end{pmatrix} \quad (113)$$

$$\mathbf{W}^3 = \begin{array}{c} \begin{array}{ccccccc} \overbrace{\phantom{\mathbf{I}_{|Q|}}}^{1:C_1} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_1:C_2} & \overbrace{\phantom{\mathbf{I}_{|A|}}}^{C_2:C_3} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_3:C_4} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_4:C_5} & \overbrace{\phantom{\mathbf{1}_{|\Gamma|}}}^{C_5+1} & \overbrace{\phantom{\mathbf{1}_{|\Gamma|}}}^{C_5+2} \end{array} \\ \begin{pmatrix} \mathbf{I}_{|Q|} & & & & & & \\ & \mathbf{I}_{|\Gamma|} & & & & -\mathbf{1}_{|\Gamma|} & -\mathbf{1}_{|\Gamma|} \\ & & \mathbf{I}_{|A|} & & & & \\ & & & \mathbf{I}_{|\Gamma|} & & \mathbf{1}_{|\Gamma|} & \\ & & & & \mathbf{I}_{|\Gamma|} & & \mathbf{1}_{|\Gamma|} \end{pmatrix} \end{array} \quad (114) \qquad \mathbf{b}^3 = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \\ -\mathbf{1}_{|\Gamma|} \\ -\mathbf{1}_{|\Gamma|} \end{pmatrix} \quad (115)$$

$$\mathbf{W}^4 = \begin{array}{c} \begin{array}{ccccc} \overbrace{\phantom{\mathbf{I}_{|Q|}}}^{1:C_1} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_1:C_2} & \overbrace{\phantom{\mathbf{I}_{|A|}}}^{C_2:C_3} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_3:C_4} & \overbrace{\phantom{\mathbf{I}_{|\Gamma|}}}^{C_4:C_5} \end{array} \\ \begin{pmatrix} \mathbf{I}_{|Q|} & & & & \\ & \mathbf{I}_{|\Gamma|} & & \mathbf{I}_{|\Gamma|} & \mathbf{I}_{|\Gamma|} \\ & & \mathbf{I}_{|A|} & & \end{pmatrix} \end{array} \quad (116) \qquad \mathbf{b}^4 = \begin{pmatrix} \mathbf{0}_{|Q|} \\ \mathbf{0}_{|\Gamma|} \\ \mathbf{0}_{|A|} \end{pmatrix} \quad (117)$$

Lastly, we set $\mathbf{W}^5$ and $\mathbf{b}^5$ such that the the function $\mathbf{x} \mapsto \mathrm{ReLU}\big(\mathbf{W}^5\mathbf{x} + \mathbf{b}^5\big)$ computes the one-hot encoding of the three input one-hot encodings $[\![q_t]\!]$, $[\![s_t]\!]$, $[\![a_{t-1}]\!]$ by implementing the logic AND operations, as in Svete and Cotterell (2024, Lemma B.1).